



UNIVERSITI PUTRA MALAYSIA
“BERILMU BERBAKTI”

CCS3700 – EMBEDDED SYSTEM PROGRAMMING
LAB 2

MORSE CODE LED

LECTURER’S NAME:
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MATRIC NO:
230405

SUBMITTED DATE:
6/4/2026

LAB 2 (WEEKS 2 & 3) – Individual

OBJECTIVE:

The main objective of this lab is to introduce the students to digital I/O conversion and transfer. Students will be able to know how to configure digital I/O for the Arduino board, and create Arduino sketch programs based on Morse Code and Traffic Light examples.

DURATION:

2 weeks/6 hours

DUE DATE OF SUBMISSION:

Submit your assignment to your demonstrator before the next lab.

Don't forget to sign the assignment submission form or your marks will not be graded.

PROGRAMME OBJECTIVE:

CPS3-3: Apply appropriate algorithms and techniques to computing solutions.

RUBRIC:

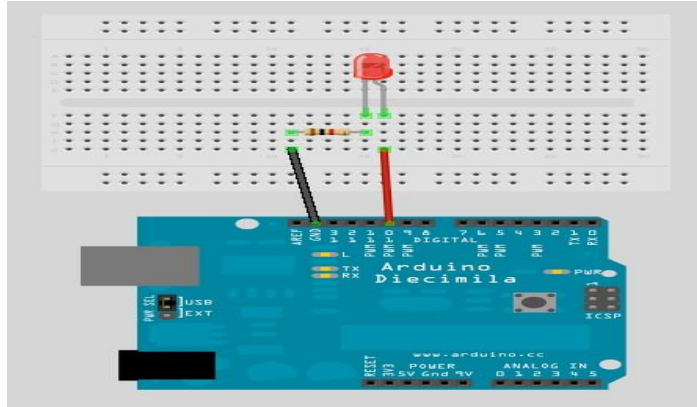
CRITERIA	LEVEL 1 Very Poor	LEVEL 2 Poor	LEVEL 3 Good	LEVEL 4 Very Good	LEVEL 5 Excellent
The ability to perform digital I/O tasks.	Unable to perform ability to perform digital I/O tasks with very limited success (< 40%).	Able to perform digital I/O tasks with limited success ($\geq 40\%$, < 60%).	Able to perform digital I/O tasks with some success ($\geq 60\%$, < 80%).	Able to perform digital I/O tasks with considerable success ($\geq 80\%$, < 100%).	Able to perform digital I/O tasks with outstanding success (100%).

Part 1: LED Flasher

In this lab exercise, students will attempt to build a simple circuit (refer Figure 1 below) that enables a LED to flash.

Hardware

Breadboard
Red LED
150 Ω resistor
Jumper wires

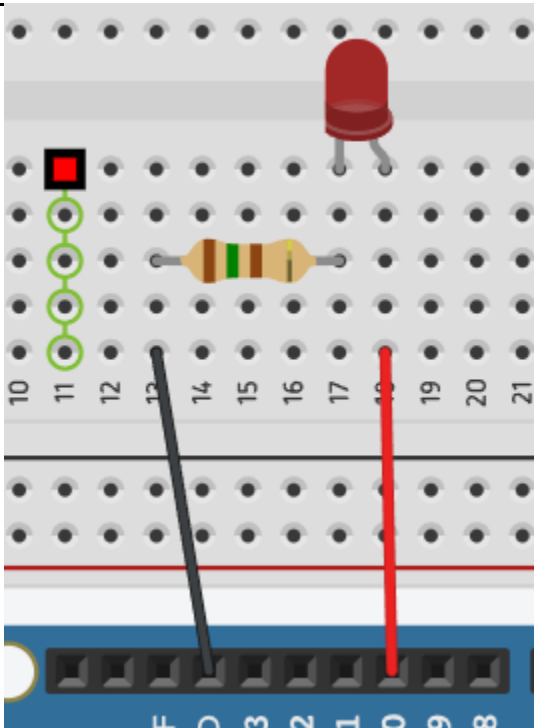


Make sure that your LED is connected the right way with the longer leg connected to Digital Pin 10. The longer leg is the anode of the LED and always must go to the +5V supply (in this case coming out of Digital Pin 10) and the shorter leg is the cathode and must go to GND (Ground). Once everything is connected, power up your Arduino via USB cable.

Instruction

Write the required program and compile it into your Arduino. If you have done everything right, you should now see the Red LED on the breadboard flashing ON and OFF every second.

ANSWER



It lights up every second

```
int LED = 10;

void setup()
{ pinMode(LED, OUTPUT); }

void loop()
{ digitalWrite(LED, HIGH);
  delay(1000);
  digitalWrite(LED, LOW);
  delay(1000); }
```

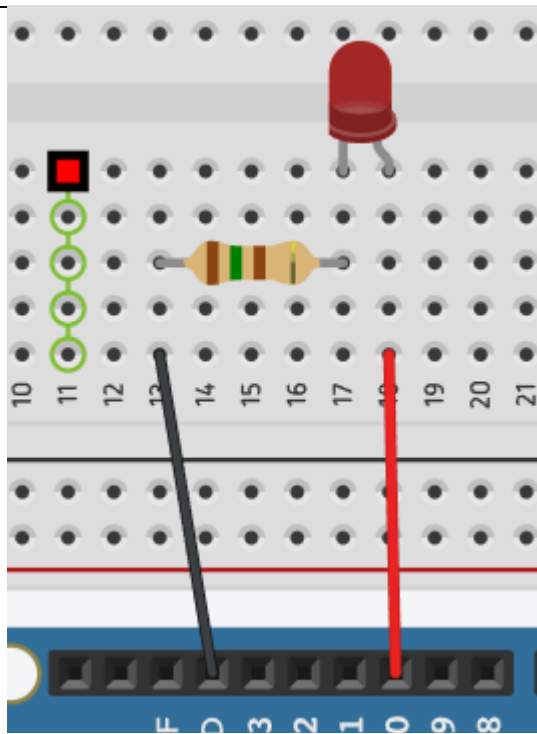
Part 2: Morse Code

In this exercise, we are going to deploy the same circuit set up as in Part 1 but will use some different code to make the LED display a message in **Morse Code**. In this case, we are going to get the LED to signal the letters S.O.S., which is the international morse code distress signal. **Morse Code** (see **Appendix**) is a type of character encoding that transmits letters and numbers using patterns of On and Off. It is therefore nicely suited to our digital system as we can turn an LED on and off in the necessary pattern to spell out a word or a series of characters. In this case we will be signaling **S.O.S.** which in the Morse Code alphabet is three dits (short flash), followed by three dahs (long flash), followed by three dits again. We can therefore now code our sketch to flash the LED on and off in this pattern, signaling SOS.

Instruction

Write the required program and compile it into your Arduino board. If all goes well, you will now see the LED flash the Morse Code SOS signal, wait 5 seconds, then repeat.

ANSWER



Same setup, different code

```
int LED = 10;
void setup() { pinMode(LED, OUTPUT); }

void loop() {
  // --- Letter 'S' (Three short flashes) ---
  for (int i = 0; i < 3; i++) {
    digitalWrite(LED, HIGH); delay(200);
    digitalWrite(LED, LOW); delay(200); }

  delay(400);

  // --- Letter 'O' (Three long flashes) ---
  for (int i = 0; i < 3; i++) {
    digitalWrite(LED, HIGH); delay(600);
    digitalWrite(LED, LOW); delay(200); }
  delay(400);

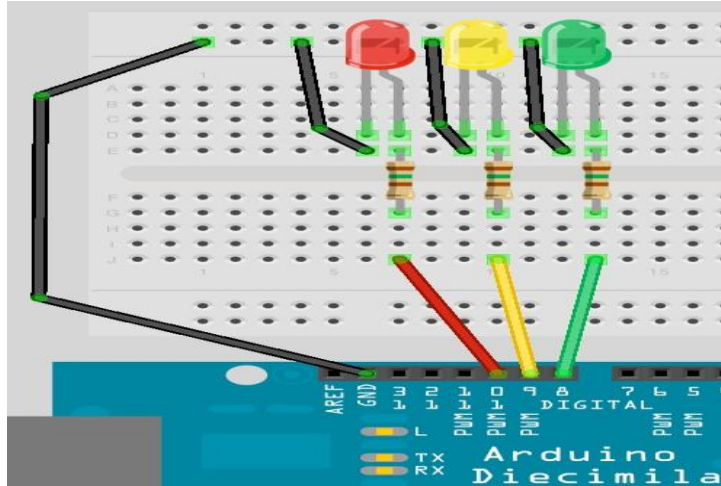
  // --- Letter 'S' (Three short flashes) ---
  for (int i = 0; i < 3; i++) {
    digitalWrite(LED, HIGH); delay(200);
    digitalWrite(LED, LOW); delay(200); }
  delay(5000); }
```

Part 3 Traffic Lights

In this lab exercise, students will create a set of traffic light systems that will change green to red, via yellow, and back again, after a set of length of time using the 4-state system.

Hardware

Breadboard
Red LED
Yellow LED
Green LED
3 x 220Ω resistors
Jumper wire

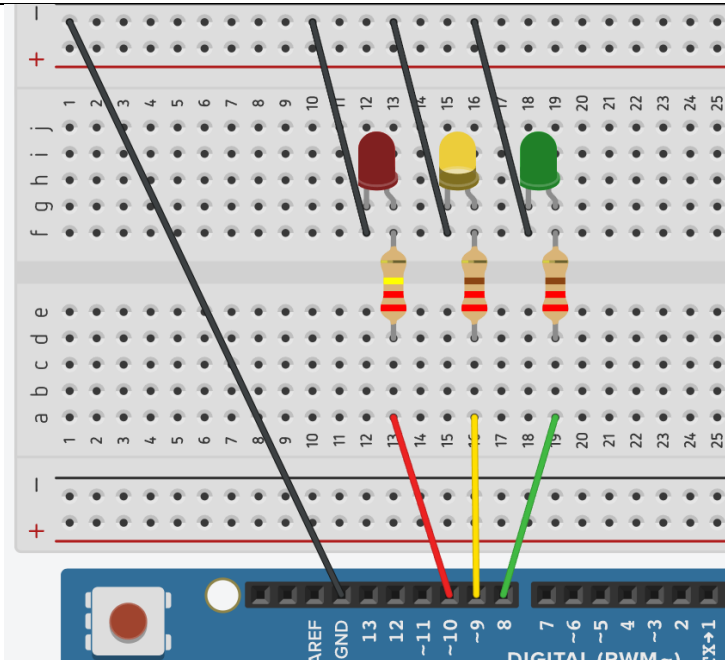


As shown in Figure 2 above, connect 3 LEDs with the anode of each going to Digital Pins 8,9, and 10, via a 220Ω resistor each. Take a jumper wire from Ground to the ground rail at the top of the breadboard and a ground wire goes from the cathode leg of each LED to the common ground rail.

Instruction

Write the required program and compile it into your Arduino board

ANSWER



```
int greenLED = 8;
int yellowLED = 9;
int redLED = 10;

void setup() {
  pinMode(greenLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(redLED, OUTPUT);
}

void loop() {
  digitalWrite(greenLED, HIGH);
  digitalWrite(yellowLED, LOW);
  digitalWrite(redLED, LOW);
  delay(5000);

  digitalWrite(greenLED, LOW);
  digitalWrite(yellowLED, HIGH);
  digitalWrite(redLED, LOW);
  delay(2000);

  digitalWrite(greenLED, LOW);
  digitalWrite(yellowLED, LOW);
  digitalWrite(redLED, HIGH);
  delay(5000);

  digitalWrite(greenLED, LOW);
  digitalWrite(yellowLED, HIGH);
  digitalWrite(redLED, LOW);
  delay(2000);
}
```